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Department of Cybersecurity

Network Administration

Linux Commands and Using ShellGPT as an AI Assistant

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Introduction

This assignment focuses on practicing Linux commands used in system and network administration and using ShellGPT as an AI assistant inside an Ubuntu virtual machine.

The first part covers Linux manual pages, file system navigation, file and directory management, viewing file contents, text processing, user password management, and text editing.

The second part demonstrates how ShellGPT can generate Linux commands from natural-language instructions. The experiments include displaying network interfaces and discovering active hosts using Nmap.

The main objectives are to understand Linux commands, practice their usage, interpret their outputs, and evaluate how an AI assistant can support network administration tasks.

2. Part One: Linux Commands

2.1. Checking the System Information

The first step was to identify the current user, working directory, and operating system version.

The following commands were used:

whoami

pwd

lsb_release -a

The whoami command displays the current username. The pwd command displays the current working directory, while lsb_release -a provides information about the Linux distribution.

The system information showed that the operating system was Ubuntu 24.10, with the codename Oracular. The message “No LSB modules are available” appeared, but the distribution information was still displayed successfully.

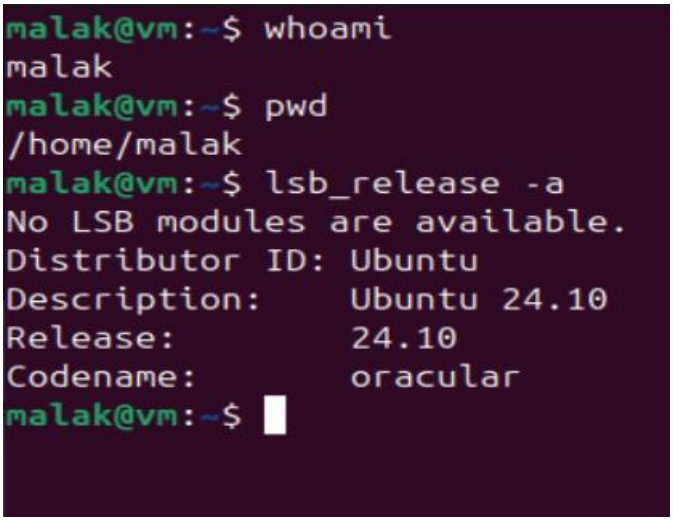


Figure 1: Checking the current user, working directory, and Ubuntu version.

2.2. Creating the Lab Directory

A separate directory was created to organize the assignment files.

The following commands were used:

```
mkdir ~/labadmin

cd ~/labadmin

pwd

ls -la
```

The mkdir command creates the directory, while cd changes the current directory. The pwd command verifies the directory location, and ls -la lists its contents, including hidden files.

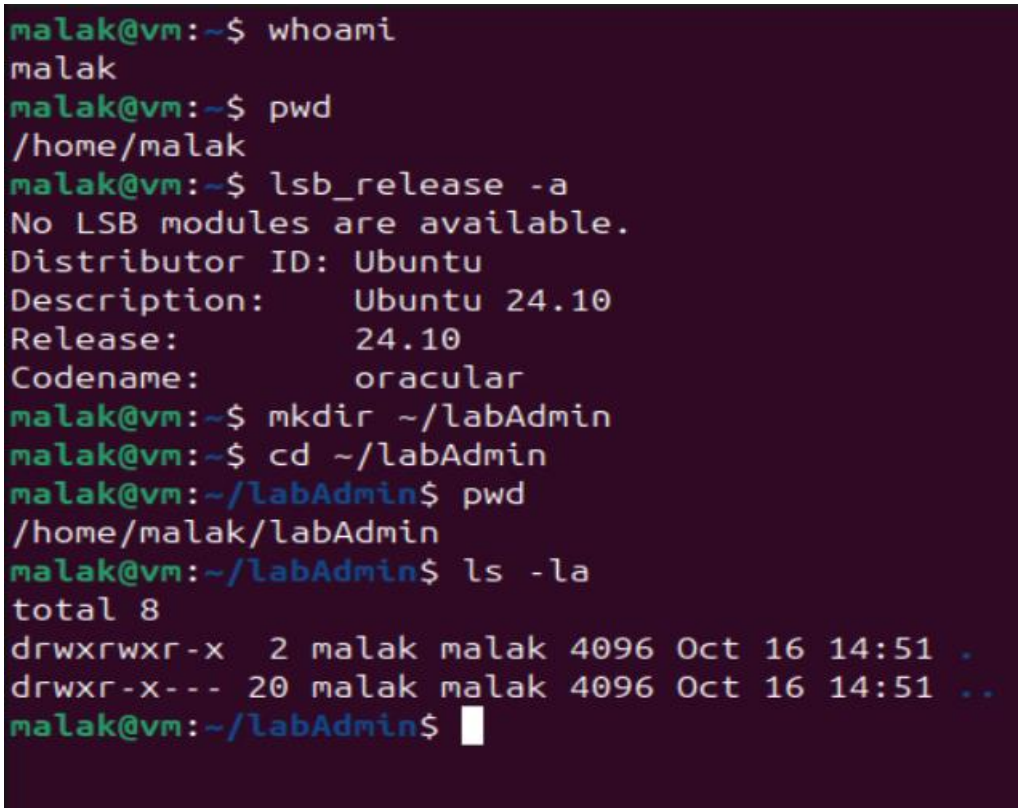


Figure 2: Creating the lab directory and checking its contents.

2.3. Getting Help with the man Command

The man command displays manual pages that explain Linux commands, their options, and their usage.

The following command was used:

man ls

The manual page explains how to use ls to list directory contents. The q key exits the manual page.

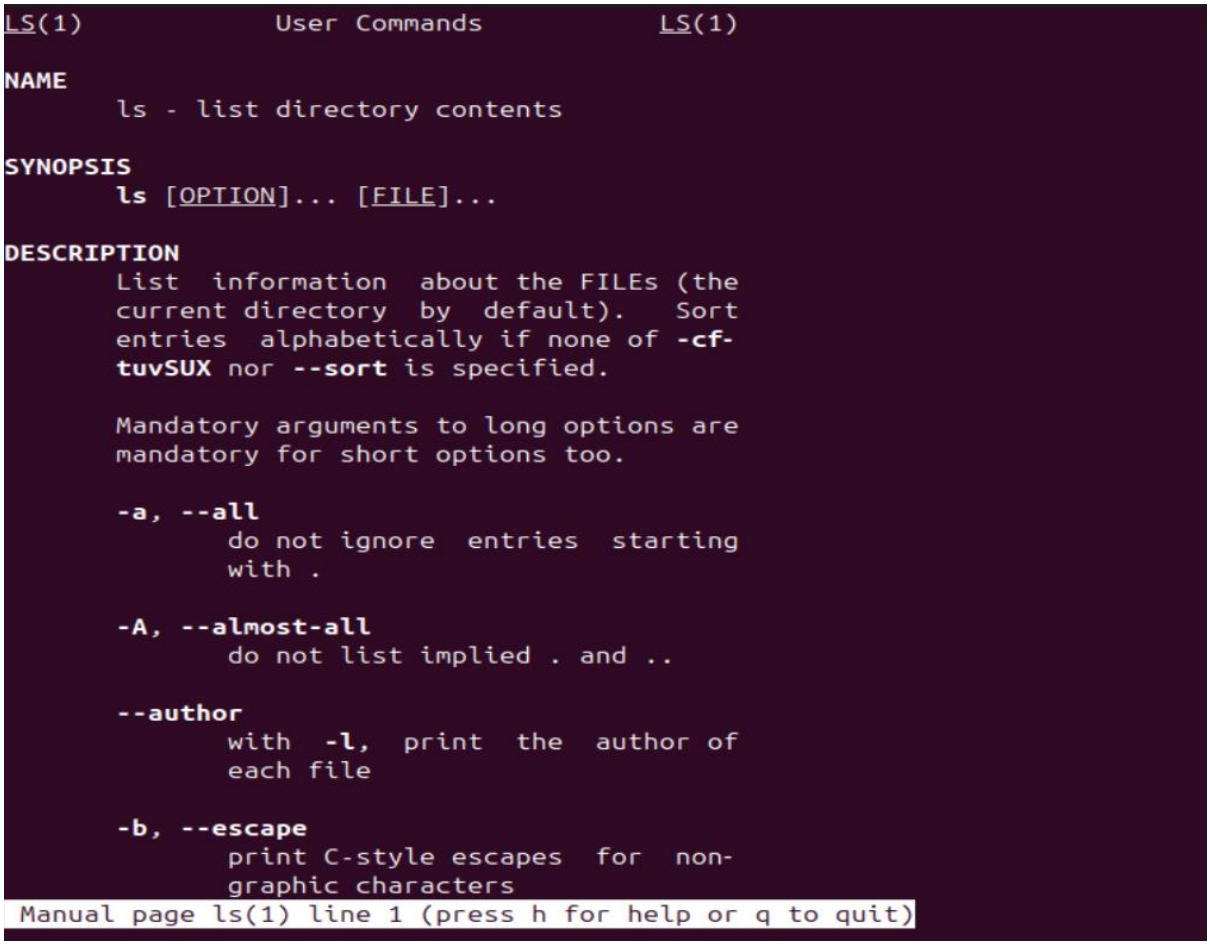


Figure 3: Viewing the manual page of the ls command.

2.4. Searching Manual Pages

The man -k command searches manual page descriptions using a keyword.

The following commands were used:

man -k network

man -k "network interface" | head -10

The first command searches for manual entries related to networks. The second searches for a longer phrase and uses a pipe to send the results to head -10, which displays the first ten lines.

```
malak@vm:~/labAdmin$ man -k "network interface" | head -10
if_freenameindex (3) - get network interface...
if_indextoname (3)   - mappings between netw...
if_nameindex (3)     - get network interface...
if_nametoindex (3)   - mappings between netw...
ifconfig (8)         - configure a network i...
nameif (8)           - name network interfac...
netqtop-bpfcc (8)    - Summarize PPS, BPS, a...
slattach (8)         - attach a network inte...
malak@vm:~/labAdmin$
```

Figure 4: Searching Linux manual pages for network-related commands.

2.5. Reading the Linux Introduction

The following command was used:

man intro

This manual page introduces Linux user commands and the command-line interface. It explains how a shell receives and processes commands entered by the user.

The q key was used to exit the manual page.

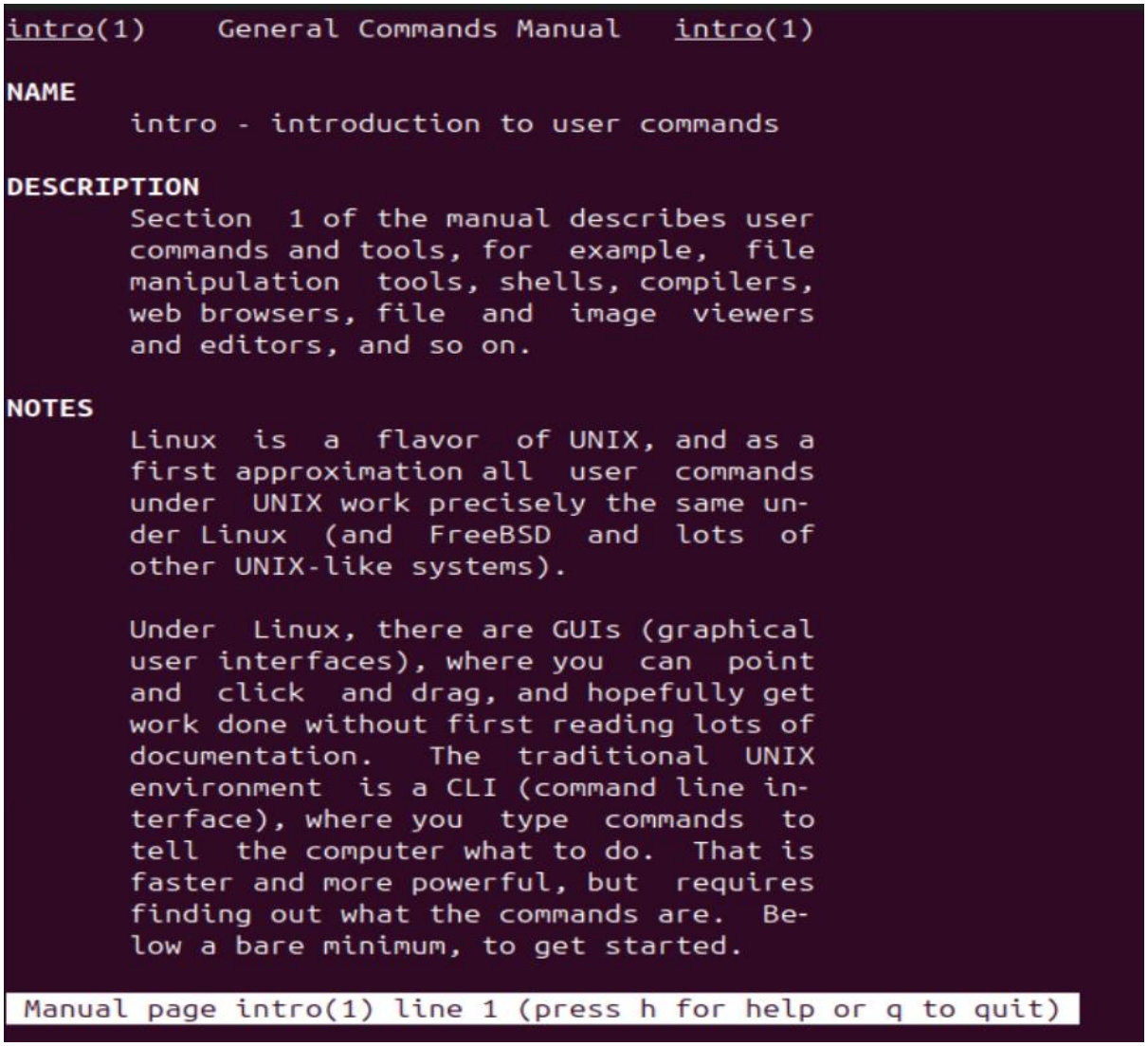


Figure 5: Reading the introduction to Linux user commands.

2.6. Navigating the File System

Linux provides commands for moving between directories and exploring the file system.

The following commands were used:

cd /usr/local/bin

pwd

cd ~

pwd

cd ..

pwd

cd ~/..

pwd

cd ~/labadmin

pwd

The cd command changes the current directory. An absolute path starts from the root directory, ~ represents the user's home directory, and .. represents the parent directory. The pwd command confirms the current location after each change.



Figure 6: Navigating directories using absolute and relative paths.

2.7. Exploring System Directories

The following commands were used to explore important Linux directories:

```
ls /
ls -ld /bin /sbin
ls /usr/bin | head -5
```

The ls / command lists the main directories in the root file system. The ls -ld command displays information about the specified directories or symbolic links. The last command displays the first five lines of the listing for /usr/bin.

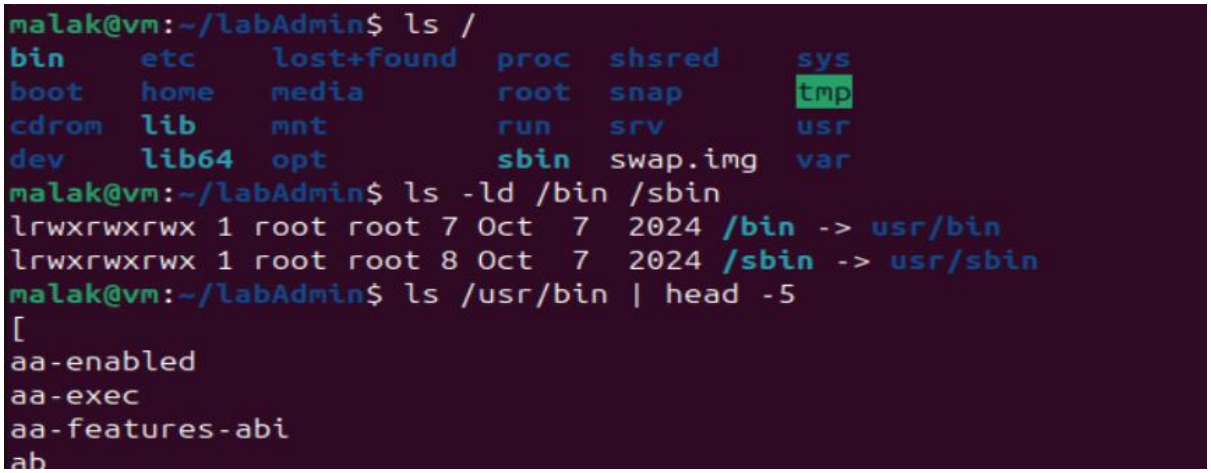


Figure 7: Exploring the root directory, system links, and available commands.

2.8. Creating, Copying, and Moving Files

A set of files and directories was created to practice basic file management.

The following commands were used:

```
cd ~/labadmin

mkdir dir1 dir2

echo "hello" > file1

cat file1

cp file1 file2

cp file1 dir1

cp -r dir1 dir3

mv file2 file3

mv file3 dir2

ls

ls dir1 dir2 dir3
```

The mkdir command creates directories, and echo writes text into a file using output redirection. The cat command displays the file contents. The cp command copies files, while cp -r copies a directory and its contents. The mv command renames or moves files.

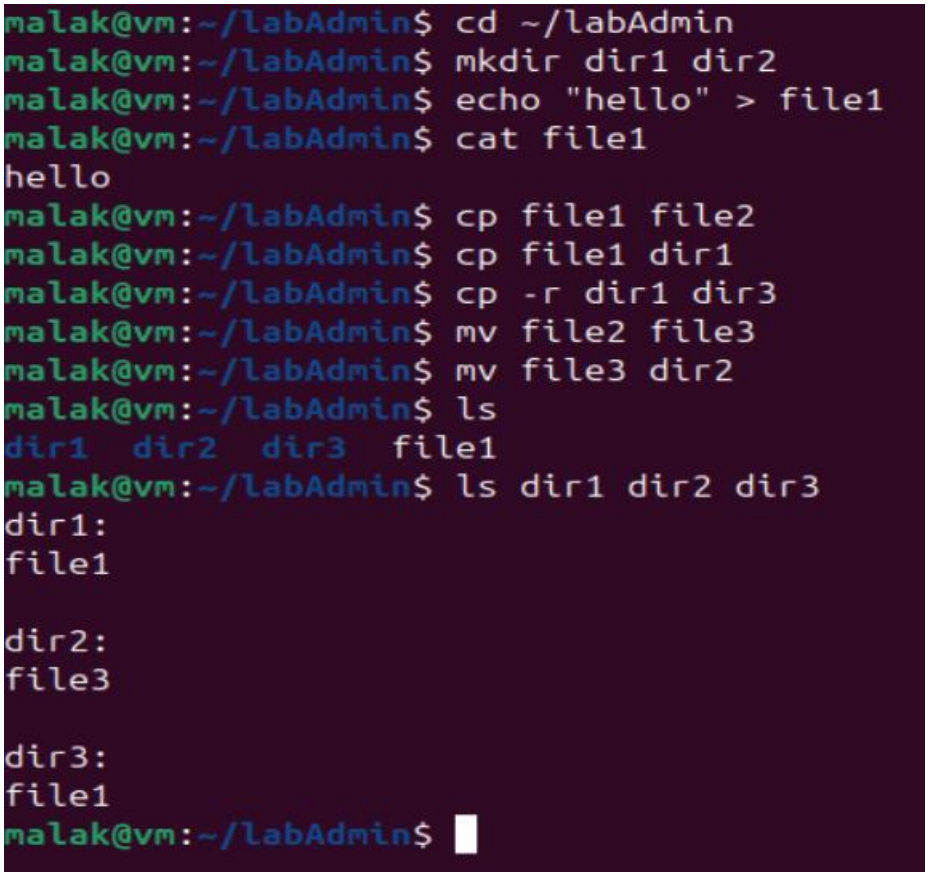


Figure 8: Creating, copying, and moving files and directories.

2.9. Deleting Files and Directories

The following commands were used to demonstrate file and directory deletion:

rm dir1/file1

rmdir dir1

rmdir dir3

The rm command deletes a file. The rmdir command removes an empty directory. The attempt to remove dir3 failed because it still contained a file.

The directory was then removed using:

rm -r dir3

ls -la

The rm -r command deletes a directory and its contents recursively. It must be used carefully because deleted files may not be recoverable.

```
malak@vm:~/labAdmin$ rm dir1/file1
malak@vm:~/labAdmin$ rmdir dir1
malak@vm:~/labAdmin$ rmdir dir3
rmdir: failed to remove 'dir3': Directory not empty
```

Figure 9: Demonstrating that rmdir cannot remove a non-empty directory.

```
malak@vm:~/labAdmin$ rm -r dir3
malak@vm:~/labAdmin$ ls -la
total 16
drwxrwxr-x  3 malak malak 4096 Oct 16 15:13 .
drwxr-x--- 20 malak malak 4096 Oct 16 14:51 ..
drwxrwxr-x  2 malak malak 4096 Oct 16 15:10 dir2
-rw-rw-r--  1 malak malak    6 Oct 16 15:09 file1
malak@vm:~/labAdmin$
```

Figure 10: Removing a non-empty directory using rm -r and checking the result.

2.10. Viewing File Contents with more and less

The following commands were used:

more /etc/services

less /etc/services

The /etc/services file contains service names and their associated port numbers and protocols.

The more command displays the file page by page. The less command also allows the user to navigate through the file and search its contents. Both commands can be exited by pressing q.

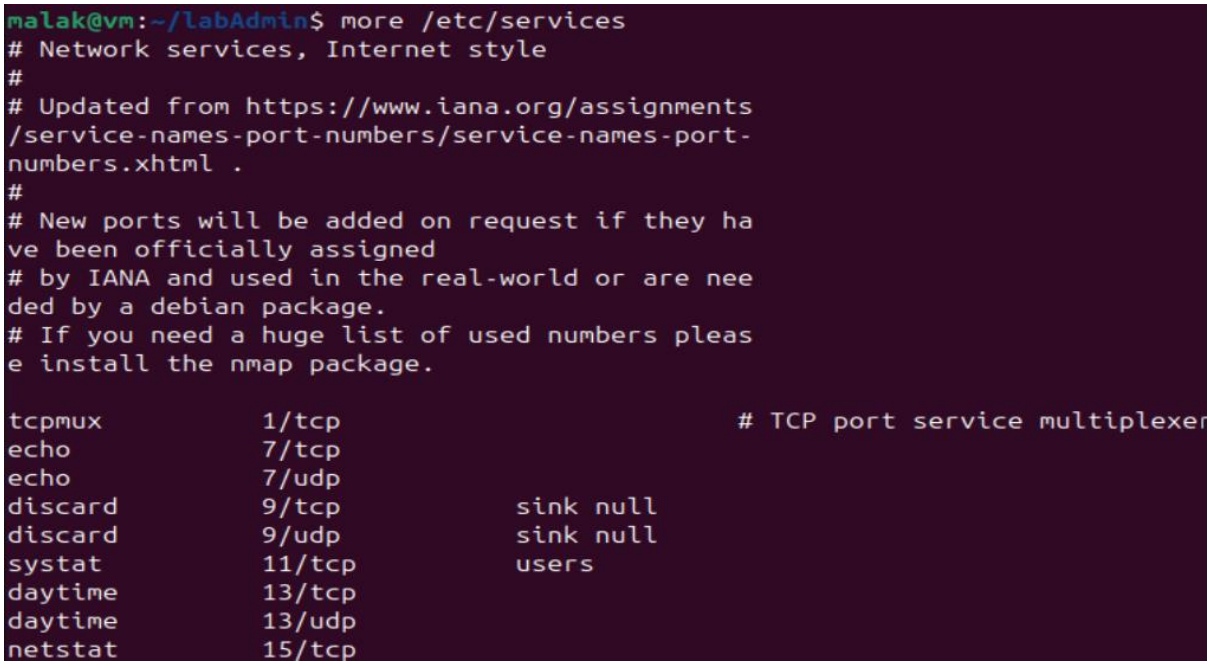


Figure 11: Viewing the /etc/services file using more.



Figure 12: Viewing the /etc/services file using less.

2.11. Using head, tail, wc, and sort

The following commands were used to inspect and process /etc/passwd:

```
head -5 /etc/passwd

tail -5 /etc/passwd

wc -l /etc/passwd

sort /etc/passwd | head -5

sort -r /etc/passwd | head -5

sort -t: -n -k3 /etc/passwd | head -5
```

The head command displays the beginning of a file, while tail displays its end. The wc -l command counts the lines. The sort command sorts the lines alphabetically, in reverse order, or numerically according to a specified field.

In the last command, -t: sets the colon as the field separator, -n enables numerical sorting, and -k3 selects the third field.

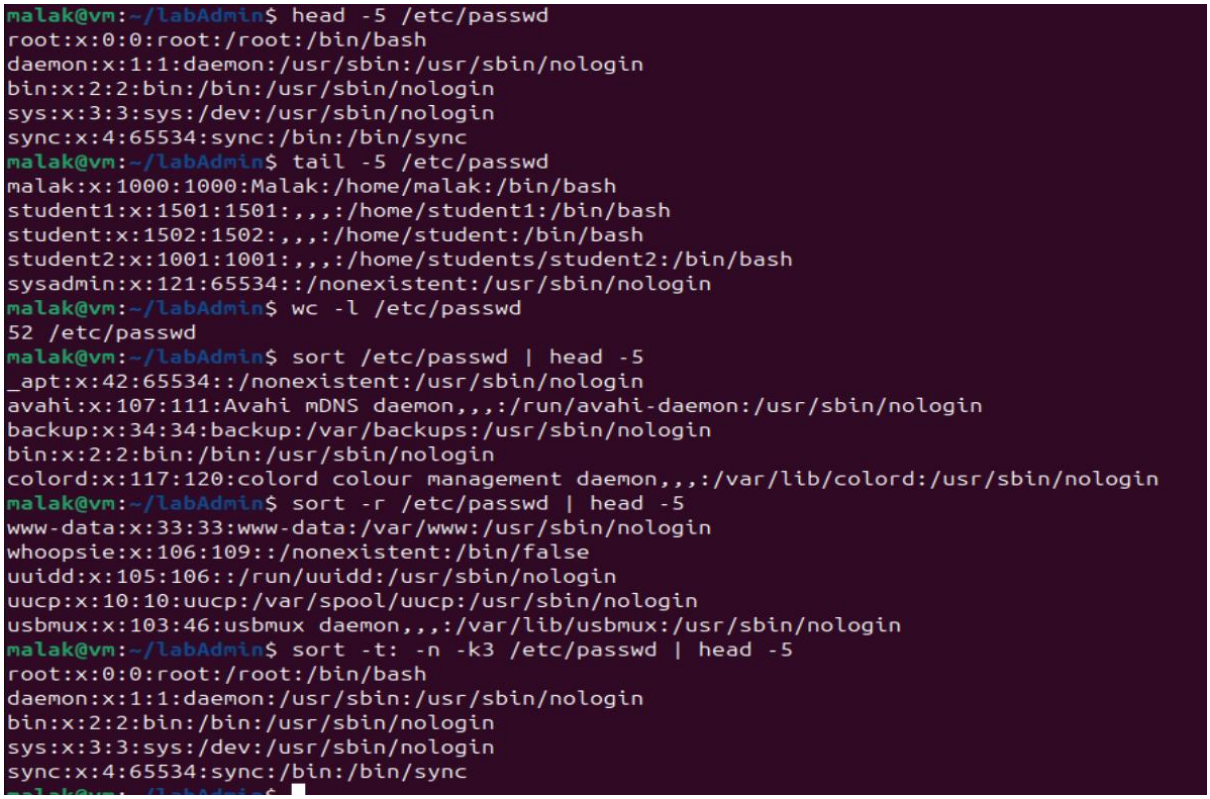


Figure 13: Using head, tail, wc, and sort to examine /etc/passwd.

2.12. Understanding the /etc/passwd File

The /etc/passwd file stores account information. Each line normally contains seven fields separated by colons.

The following commands were used:

head -5 /etc/passwd

grep "^$(whoami):" /etc/passwd

The first command displays the first five account entries. The second searches for the entry associated with the current user.

The seven fields represent the username, password placeholder, user ID (UID), group ID (GID), comment, home directory, and login shell.

The password field commonly contains x, indicating that the protected password information is stored in /etc/shadow.

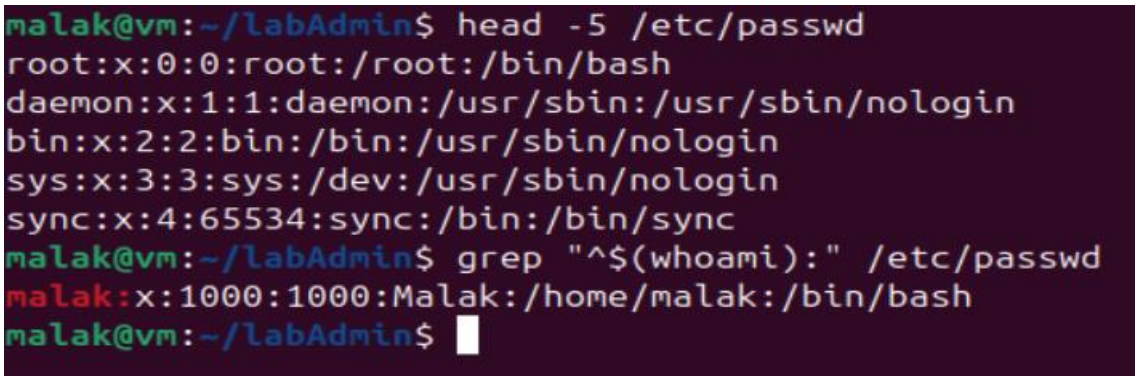


Figure 14: Examining account entries and identifying the current user.

2.13. Text Processing Commands

Several Linux commands were used to extract, search, combine, and transform text.

The following commands were used:

```
cut -d: -f1 /etc/passwd | head -5

cut -d: -f1 /etc/passwd | sort | uniq | head -5

grep bash /etc/passwd
```

The cut command extracts fields from each line. The sort and uniq commands sort entries and remove adjacent duplicate lines. The grep command searches for lines containing a specified pattern.

The following commands save usernames and login shells into separate files and combine them:

```
cut -d: -f1 /etc/passwd > a.txt

cut -d: -f7 /etc/passwd > b.txt

paste a.txt b.txt | head -5
```

Finally, the tr command was used to replace the letter o with the digit 0:

```
echo "hello world" | tr o 0
```

The expected output is hell0 w0rld.



Figure 15: Extracting, sorting, and searching text using cut, sort, uniq, and grep.

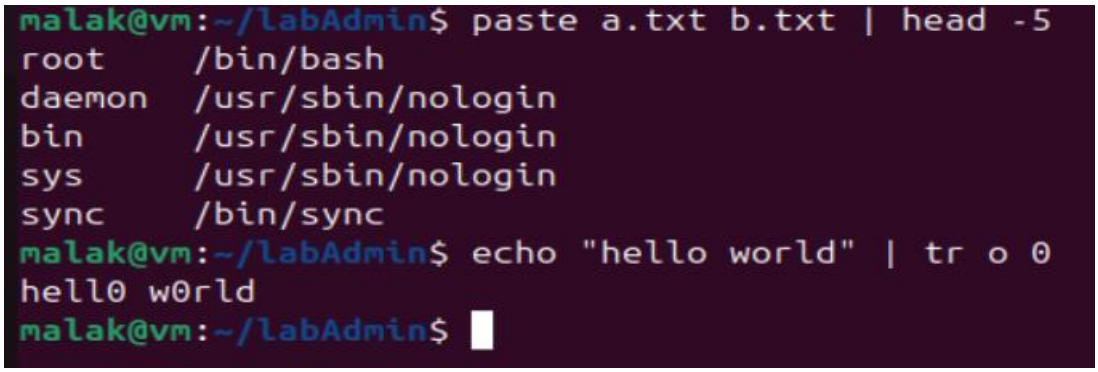


Figure 16: Combining text files using paste and replacing characters using tr.

2.14. Managing User Passwords

The passwd command is used to change the password of the current user.

passwd

The command may request the current password and then prompt for a new password. Password characters are not displayed while typing, which is a security feature.

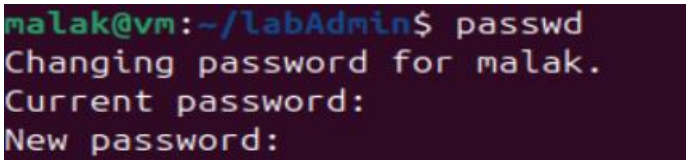


Figure 17: Using the passwd command.

2.15. Editing a File with vi

The vi editor can be used to create and modify text files.

The following command opens a file:

cd ~/labadmin

vi -N test.txt

Inside the editor, the i key enters Insert Mode. The text Hello from vi was entered, and the Esc key returned the editor to command mode. The :wq command saves the file and exits the editor.

The file was then checked using:

cat test.txt

The displayed text confirms that the content was saved successfully.

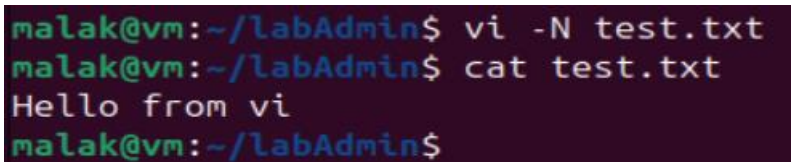


Figure 19: Verifying the saved file using cat.

3. Part Two: Using ShellGPT as an AI Assistant

3.1. Introduction to ShellGPT

ShellGPT is a command-line AI assistant that generates shell commands from natural-language instructions. It allows users to describe a task in English and review the generated command before executing it.

In this assignment, ShellGPT was used with the local Llama 3.2 model through Ollama.

When ShellGPT generates a command in shell mode, it presents the following choices:

* E — Execute: Run the generated command.

* M — Modify: Edit the command.

* D — Describe: Explain the command.

* A — Abort: Cancel the operation.

The user should review the command before execution to ensure it is appropriate for the task.

3.2. Showing Network Interfaces

ShellGPT was asked to display the network interfaces using the following command:

sgpt --shell "Show network interfaces using ip"

The AI agent generated the command:

ip link show

This command displays information about the network interfaces available on the Linux system, including their names and link states.

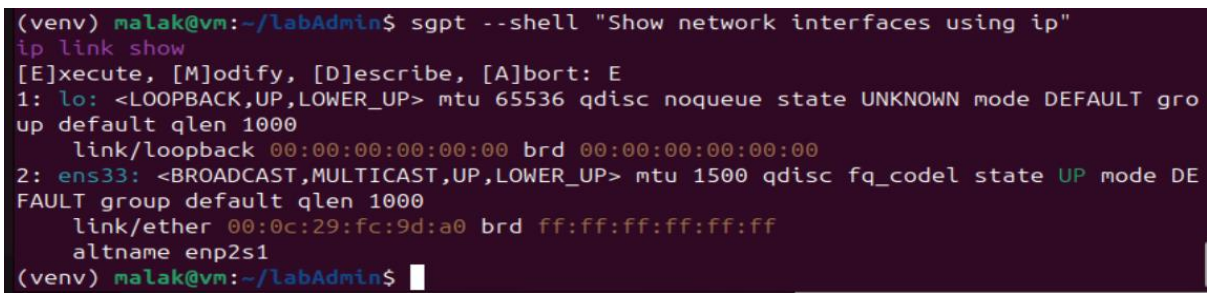


Figure 20: Using ShellGPT to generate and execute the ip link show command.

3.3. Discovering Active Hosts Using Nmap

ShellGPT was asked to generate a network discovery command using the following prompt:

sgpt --shell "Use nmap to discover live hosts on my local network. First identify my subnet, then generate a ping scan command."

The AI agent generated the following command:

nmap -sP 192.168.1.0/24

After reviewing the command, the execution option was selected.

Nmap produced the following output:

Starting Nmap 7.95 (https://nmap.org) at 2026-10-16 19:57 EEST

Nmap scan report for 192.168.1.1

Host is up (0.011s latency).

Nmap scan report for 192.168.1.115

Host is up (0.11s latency).

Nmap scan report for 192.168.1.255

Host is up (0.0044s latency).

Nmap done: 256 IP addresses (3 hosts up) scanned in 4.64 seconds

The scan reported three responsive IP addresses:

IP address	Result
192.168.1.1	Reported as up
192.168.1.115	Reported as up
192.168.1.255	Reported as up

The last address is commonly the broadcast address for a /24 network and should not automatically be interpreted as a separate physical device. The actual network range should be verified against the VM's network configuration before the results are interpreted as a complete description of the local network.

The -sP option is an older form of Nmap's host-discovery option. The modern equivalent is -sn, which performs host discovery without a port scan.

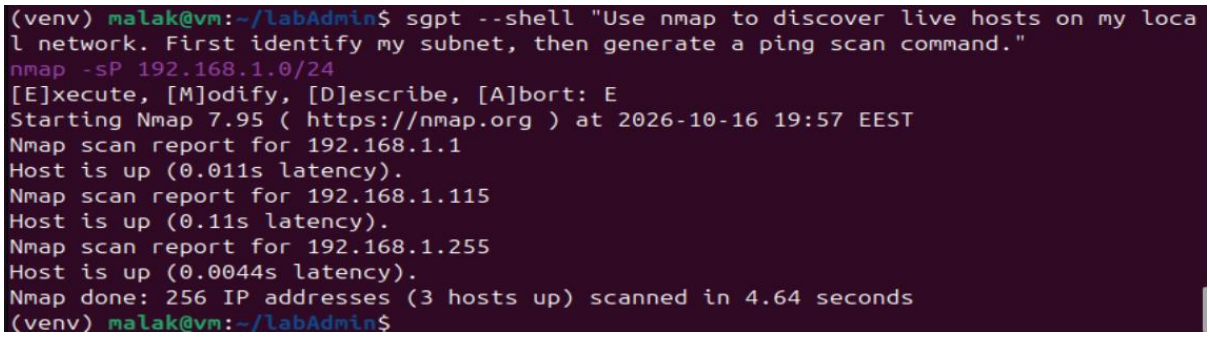


Figure 21: Using ShellGPT to generate and execute the Nmap host-discovery command.

3.4. Comparing Manual Commands with an AI Agent

The experiments demonstrate two ways to work with Linux commands.

	Typing commands myself	Using the AI agent
Input	Exact command and options	A sentence in plain English
Speed	Fast if I remember the command	Fast even for complex tasks
Learning	Builds memory of the syntax	Can teach through [D]escribe
Risk	My own typing mistakes	Wrong or unsafe command if I execute blindly
Control	Full	I must review before pressing E

The AI assistant can simplify command generation, but its output must be checked before it is executed.

4. Conclusion

This assignment provided practical experience with Linux commands used in system and network administration. The exercises covered manual pages, directory navigation, file management, file viewing, text processing, user password management, and editing files using vi.

The second part demonstrated how ShellGPT can translate natural-language requests into Linux commands. The network interface experiment used ip link show, while the Nmap experiment demonstrated host discovery on a specified network range.

Overall, the assignment showed that AI assistants can support network administration tasks, but administrators must understand the commands they execute and verify that the commands and target networks are appropriate.